INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

ACCELEROMETER, AIRCRAFT RANGE -5 TO 10G UNITS, INTEGRALLY LIGHTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers integrally lighted -5 to +10g range, aircraft accelerometers.

1.2 Classification. Accelerometers covered by this specification shall be of the following types, as specified (see 6.2):

ABU-4A/A - Nonhermetically sealed
ABU-4B/A - Hermetically sealed

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-P-116 Preservation, Methods Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to; Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610
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MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, For Instrument Cover Glasses And Lighting Wedges
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge And Cover
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-785	Reliability Program For Systems And Equipment Development And Production
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MS23009	Accelerometer-Aircraft, Integrally Lighted, ABU-4A/A And ABU-4B/A
MS28105	Cover Glass, Aircraft Instrument Dial
MS33558	Numerals And letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instruments

HANDBOOK

MIL-HDBK-781	Reliability Test methods, Plans And Environments For Engineering Development, Qualification And Production

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building 4D, Philadelphia, PA 19111-5094)

2.2.2 Other government document, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Air Force Drawing 8346561

(A copy of the Air Force drawing is available from OC-ALC/TILDO, Tinker AFB OK 73145-5990).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 MS sheets. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheet. In the event of any conflict between the requirements of this specification and the MS sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Materials

3.3.1 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to occur in storage or normal service.

3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used, except where magnetic materials are essential.

3.3.3 Lubricants. When required, the lubricant used shall be in accordance with MIL-L-6085. The amount of lubricant used shall be sufficient for at least 1,000 hours of operation.

3.3.4 Protective treatment. When materials are used in the construction of the accelerometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extreme of climatic and environmental conditions shall be avoided.

3.4. Design and construction

3.4.1 Design. The accelerometer shall be designed to indicate aircraft accelerations in maneuvers and rough air, and shall conform to MS23009 and FIGURE 1.

3.4.2 Response. The accelerometer shall be designed to comply with the response characteristics specified in 4.5.4.

3.4.3 Construction. The construction shall be reasonably simple to permit overhaul or repair with the use of tools and fixtures normally available commercially. The accelerometer shall be so constructed as to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.4.4 Pivots, bearings and gears. Pivots, bearings, and gears shall operate smoothly and shall be as near frictionless as practical.

3.4.5 Case. The case shall conform to MS23009. The Bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein.

3.4.5.1 Hermetic sealing (ABU-4B/A). The case shall be hermetically sealed and shall be so designed that the internal mechanism may be removed, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be such that the sealing will not be affected by the action of any atmosphere to which the indicator may be subjected. The case shall meet the necessary pressure requirements when tested for leaks by means of a mass spectrometer-type of helium leak detector.

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3.4.5.2 Filling medium (ABU-4B/A). The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.4.5.3 Filling medium tube. The filling medium tube for the hermetically sealed unit shall be located on the rear of the case and shall not interfere with the locking knob operation or the electrical connection.

3.4.6 Cover glass. The cover glass shall conform to MS28105 and MIL-G-81704.

3.4.6.1 Reflection-reducing coating. All cover glass and lighting wedges shall be provided with a reflection reducing coating that meets the requirements of MIL-C-14806. This requirement can be accomplished by obtaining a certificate of compliance from the manufacturer of the anti-reflective coating that states the coating meets the requirements of MIL-C-14806.

3.4.7 Dial. The dial shall conform to FIGURE 1 and shall be fastened securely to the case or upon the frame of the mechanism by not less than two screws in such manner that it will not loosen or turn when the accelerometer is vibrated.

3.4.7.1 Distance from dial to cover glass. The distance between dial and cover glass shall be not greater than 0.250 inch.

3.4.8 Pointers. The accelerometer shall have three pointers; one main pointer for indicating instantaneous acceleration, and two auxiliary pointers, one for indicating maximum positive acceleration, and one for indicating maximum negative acceleration. The pointers shall conform to FIGURE 1. They shall be light and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.4.9 Ratchet mechanism. Each auxiliary pointer shall be provided with a ratchet device, in order that indication of the maximum plus and minus deflections of the main pointer will be maintained until the auxiliary pointers are reset to obtain a new reading.

3.4.9.1 Auxiliary pointer return mechanism. A mechanism shall be provided for resetting the auxiliary pointers into juxtaposition with the main pointer. A knob, located as shown on MS23009, shall be provided for operating the return mechanism. The shaft of this knob shall be capable of withstanding a load of 15 ± 1 pound applied perpendicular to the shaft at the extreme outward end of the knob without permanent deformation. Stops shall be provided in order that the auxiliary pointers, when reset, shall not indicate a lower or higher reading than that indicated by the main pointer. Stops shall be a part of the mechanism connecting the pointers and shall not directly engage them. Operation of the return mechanism shall require 3 to 8 pounds of force on the knob.

3.4.9.2 Main pointer mechanism. The main power mechanism shall not engage the mechanism of the auxiliary pointers when the imposed acceleration is less than the acceleration indicated by the auxiliary pointers.

3.4.10 Stops for acceleration-responsible element. Mechanical stops shall be provided for arresting the motion of the acceleration-responsive element and the pointers when the accelerometer is subjected to accelerations beyond the limits specified. One stop shall engage the element at an acceleration of not less than -5.1g and not more than -5.5g, and the other at an acceleration of not less than +10.1g and not more than +10.5g.

3.4.11 Locking device. A device for manually locking the position of the acceleration-responsive element shall be provided in order to protect the mechanism of the accelerometer from possible damage during shipment. The device shall be controlled by means of a suitable knurled knob, located at the rear of the case and marked as shown on MS23009. The force required to lock and unlock the unit shall be 4.5± 1 pound.

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3.4.12 Adjustment. The mechanism shall be provided with means to adjust or correct the indications. This means of adjustment shall be reasonably simple to manipulate and of a type that adjustment can be accomplished with tools ordinarily possessed by an instrument repairman.

3.4.13 Indication and range. The accelerometer shall indicate vertical accelerations when the plane of the mounting lugs is vertical and the +5.0 graduation is at top center. When in this position, normal gravity shall be indicated at +1g. The accelerometer shall be designed to indicate accelerations in the range -5 to +10g.

3.4.14 Integral lighting. The indicator shall incorporate two separate parallel integral lighting circuits. The red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. In the event of conflict between this specification and either MIL-L-25467 or MIL-L-27160, this specification shall govern. Lighting shall operate on 5V ac or dc. Lighting acceptability of production indicators shall be based on visual comparison with a prime standard. The method of lighting shall be approved by the procuring activity prior to fabrication.

3.4.14.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas covering the entire face of the indicator shall be measured in foot-lamberts for brightness. A minimum of 10 areas covering the entire face of the indicator shall be measured for color. The prime standard instrument and its recorded coverall brightness and color measurements shall be approved by the procuring activity. The standard shall be checked at least every 50 hours of operation or 120 days, whichever occurs first, to determine any change in brightness or color. Variations greater than ±20 percent in two or more areas shall require return of the standard for corrective action. The data and, if required by the procuring activity, the prime standard shall be submitted for evaluation and reapproval or corrective action, or both.

3.4.15 Reliability

3.4.15.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785, task 101.

3.4.15.2 Reliability in mean time between failure (MTBF). The accelerometer shall have a minimum of 1,000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.6.

3.4.15.3 Longevity. The accelerometer shall have a total operating life of 3,000 hours.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.6 Performance. The accelerometer shall be capable of passing the tests specified in Section 4 when subjected to the following conditions:

a. Temperature ranging from +71°C to -65°C.

b. Relative humidity up to 100 percent.

c. Absolute pressures ranging from 30 inches mercury (Hg) down to 1.31 inches Hg approximately 70,000 feet altitude.

d. Fungus growth as encountered in tropical climates.

e. Exposure to salt sea atmosphere for a period of 50 hours.

f. Temperature shock in accordance with MIL-STD-810 method 503.2 procedure I.

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g. When vibrated at a double amplitude (see 6.4.1) of 0.018 to 0.020 inch and get a frequency variation between 500 and 3000 cycles per minute (cpm) for a total of 3 hours, the main pointer shall not exceed 0.2g to 0.6g range and the auxiliary pointers shall not change more than 0.2.g.

h. Static scale error shall not exceed 0.25g when subjected to the test specified in 4.6.10.

i. Static scale error and friction shall not exceed 0.25g as specified when tested in accordance with 4.6.2.

j. Dynamic scale error shall not exceed 0.25g at any specified test point.

k. With the accelerometer revolved about a magnetic compass with a magnetic field intensity of 0.17 to 0.19 oersted, the compass shall not deflect more than 1 degree.

l. The mounting lugs shall withstand a load of 175 pounds without fracture.

m. Response characteristics shall be within the limits specified in 4.5.4 when the accelerometer is vibrated at frequencies from 5 to 20 Hz at maximum g forces for each test point.

3.7 Markings. All markings shall be durable to withstand the usage encountered in service. The style and proportions of numerals and letters except the letter g, placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.7.1 Lusterless white material. The following markings shall be finished with lusterless white material, Color No. 37875 of FED-STD-595. The dimensions of the markings, when applicable shall be as specified in TABLE I.

3.7.2 Durable lusterless black. The markings ABU-4A/A or ABU-4B/A, the background of the dial, and all other markings not otherwise specified, shall be finished in durable lusterless black, color No. 37038 of FED-STD-595. The height of the markings shall be 0.062 ±0.016 inch.

3.8 Weight. Weight of each accelerometer shall not exceed 1.50 pounds.

3.9 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

TABLE I. Dimensions and markings.

Markings	Height or length (inch ±0.016)	Width of line or graduation (inch ±0.005)
Numerals	0.188	MS33558
Lettering "g Units" (6.4.2) and "Acceleration"	0.062	MS33558
Major graduations (except +1 "g"	See figure 1	0.031
+1 "g" graduations	See figure 1	0.062
Minor graduations	See figure 1	0.015
Arc-section dial	---	0.015
Knob markings	0.062	MS33558

3.10 Finishes and protective coatings

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3.10.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Dials, small holes, and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

d. Castings, containing non-aluminum-alloy integral inserts, may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films conforming to MIL-C-5541 shall not be used in lieu of anodizing.

3.10.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum where practical, in accordance with MIL-C-83488 Type I or II as applicable, and of a class that is adequate to achieve the degree of protection required. Other protective coatings, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.11 Envelope. An envelope, furnished by the contractor, shall contain and be packaged with each instrument and shall be marked as follows:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTALLATION INSTRUCTIONS,
MOUNTING SCREWS, AND REMOVABLE SPRING NUTS"

3.12 Workmanship. The accelerometer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.12.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practice. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the accelerometer, they shall be held or limited accordingly.

3.12.2 Screw assemblies. Assembly screws shall be tight. The word, tight, means that they cannot be appreciably tightened further without damage or injury to the screw or threads.

3.12.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.12.4 Soldering. Soldering shall be performed in accordance with MIL-STD-454.

3.12.5 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.12.6 Cleaning. The accelerometer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Resin flash that may crumble shall be removed.

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3.13 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.2.1.

4.2.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inch Hg and at room temperature, approximately 25°C). Unless otherwise specified, the accelerometer shall be bested without correction for temperature or atmospheric pressure variations.

4.2.2 Instrument vibration. Unless otherwise specified, the accelerometer shall be vibrated before a test reading is taken. This vibration shall be accomplished utilizing an electric motor and an attached counterweight as defined by Air Force Drawing 8346561.

4.2.3 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cpm, and shall subject the accelerometer to such vibration that a point on the case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle the diameter of which is equal to the amplitude specified herein.

4.2.4 Calibration procedure. The contractor shall furnish a drawing of the mechanism and case showing clearly the vertical distance of the center of mass of the actuating mechanism from the center of the instrument dial for each of the required test points. This drawing shall also include a table of the revolutions per minute (rpm) required for each test point computed on the basis of a rotating are whose radium to the center of the instrument dial is 20 inches. The rpm required for each test point shall be computed by the formula:

$$N = 9.5493 \quad \frac{32.2g}{r}$$

Where: N = rpm of rotating arm
g = test point
r = radius of arm in feet, corrected for mass position

4.2.4.1 Test points. N shall be computed for all test points from -5g to +10g in increments of 1g.

4.2.5 Allowable centrifuge vibration. The centrifuge utilized for accomplishing the tests herein shall not have an inherent vibration of more than 0.04q between 2 and 200 cps where "g" is defined a $g=0.0511\ f^2D$ with "D" being displacement in inches (double amplitude) and "f" is the frequency in cycles per second. Inherent vibration shall be defined as the vibration sensed by the accelerometer which is induced by the centrifuge when testing in accordance with this document. Installed on the centrifuge shall be a centrifuge rotor per Air Force Drawing 8346561 for mounting the accelerometer for test.

4.3 First article inspection.

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4.3.1 Test samples. The test samples shall consist of nine accelerometers representative of the production equipment and shall be tested in a place and manner designated in the contract, purchase order, or invitation for bids (see 6.2). Three of the samples shall be subjected to and shall meet all the tests specified under 4.5. The remaining six accelerometers shall be subjected to and shall meet the qualification reliability and longevity requirements of 4.6.1 and 4.6.3.

4.3.2 First article inspections. The first article inspections shall consist of all the tests specified under 4.5, and the first article phase of 4.6.

4.4 Conformance inspections. The conformance inspections shall consist of:

a. Individual tests	See 4.4.2
b. Sampling plan and tests	See 4.4.2

4.4.1 Individual tests. Each accelerometer shall be subjected to the following tests as described under 4.5:

a. Examination of products	See 4.5.1
b Static scale error and friction	See 4.5.2
c. Dynamic scale error	See 4.5.2.1
d. Individual test of MIL-L-25467 and MIL-L-27160	See 4.5.3
e. Return mechanism operation	See 4.5.5
f. Locking mechanism operation	See 4.5.5.3
g. Leakage (ABU-4B/A only)	See 4.5.19
h. Fogging (ABU-4B/A only)	See 4.5.20

4.4.2 Sampling plans and tests

4.4.2.1 Sampling plan A. Three accelerometers shall be selected at random from each 200 or less produced on the contract or order and subjected to the following as described under 4.4.1 and 4.5

a. Individual tests	See 4.4.1
b. Response characteristics	See 4.5.4
c. Force on knob	See 4.5.5.2
d. Low-temperature operation	See 4.5.6
e. High-temperature operation	See 4.5.7
f. Magnetic effect	See 4.5.8
g. Vibration error	See 4.5.9

h. Sampling plan A tests of MIL-L-25467 and MIL-L-27160.

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i. Locking mechanism operation ..... See 4.5.5.1

4.4.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three out of the first 15 and one item out of each additional 100 or less produced on contract or order shall be selected at random and subjected to the following tests as described under 4.4.2.1 and 4.5.

a. Sampling plan A tests ..... See 4.4.1

b. Vibration endurance ..... See 4.5.10

c. Shock ..... See 4.5.11

d. Mounting lug ..... See 4.5.12

e. High temperature exposure ..... See 4.5.13

f. Low temperature exposure ..... See 4.5.14

g. Temperature shock ..... See 4.5.15

h. Humidity ..... See 4.5.16

i. Salt fog ..... See 4.5.17

j. Fungus resistance ..... See 4.5.18

k. Sampling plan B tests of MIL-L-25467 and MIL-L-27160

l. Internal examination ..... See 4.5.21

4.4.2.3 Sampling plan C. Unless otherwise specified (see 6.2), three accelerometers shall be selected at random from each 100 or less on the contract or order and shall be subjected to the following tests.

a. Individual tests ..... See 4.4.1

b. Production reliability test ..... See 4.6.2

4.4.3 Rejection. Items failing to conform to this specification shall be rejected.

4.5 Test methods

4.5.1 Examination of product. Each accelerometer shall be examined externally to determine conformance to this specification with respect to materials, markings, and workmanship (3.3, 3.8, and 3.13, respectively).

4.5.2 Static scale error and friction. The static scale error shall be observed with the accelerometer held in each of the following positions:

a. Plane of dial vertical with +5g graduation at top (scale reading shall be +1g).

b. Plane of dial horizontal (scale reading shall be zero g).

c. Plane of dial vertical with +5g graduation at bottom (scale reading shall be -1g).

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In each position, the main pointer reading obtained shall be due entirely to gravity and not to motion in change of position of the accelerometer. The static scale error shall not exceed ±0.25g units in any of the three positions tested. Friction shall not exceed ±0.25g units at the a. (1g) and c. (-1g) test positions and ±0.3g units at the b. (0g) test position with the memory pointers engaged.

4.5.2.1 Dynamic scale error and friction. The accelerometer shall be mounted on a suitable centrifuge and centrifuge rotor per Air Force Drawing 8346561 and tested for dynamic scale error and friction for accelerations of -5, -4, -3, -2, +2, +3, +4, +5, +6, +7, +8, +9 and +10 at 1g increments for a total of 13 test points. For the first test point and without instrument vibration, the centrifuge shall be brought up to, but not exceeding, the rotation rate equivalent to the "g" level being tested. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. After re-zeroing the memory pointer, the procedure shall be repeated except that instrument vibration shall be applied per paragraph 4.2.2. The centrifuge rate shall be reduced to zero and the accelerometer reading taken from the memory pointer. The memory pointer shall then be zeroed. The reading with and without vibration shall not differ from the centrifuge input "g" reading by more than ±0.25g. This procedure shall be repeated for the remaining 12 test points. The centrifuge rate need not be returned to zero and the memory pointer reset, if capability exists to read the instruments without reducing the centrifuge's rotational rate.

4.5.3 Lighting. The indicator shall be subjected to the individual lighting tests of MIL-L-25467 and MIL-L-27160.

4.5.3.1 Visual comparison. This test shall be conducted by qualified personnel only after they have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark adapted. The prime standard and the indicator under test shall be placed side-by-side at eye level in completely dark surroundings and compared as described in the following paragraphs.

4.5.3.1.1 Brightness. With 2.50±0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminals of the prime standard, like areas of the pointers, numerals, indicia, markings, et cetera, of the indicator under test and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness settings of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar in any individual area or the lighting of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this test, they shall be rejected and returned for rework. First Article units shall not be tested by use of the prime standard.

4.5.3.1.2 Color. The color comparison test shall be conducted as specified in 4.5.3.1.1, except 4.50V shall be applied to both the production indicator and the prime standard. The two indicators shall be identical in color.

4.5.4 Response characteristics. The accelerometer shall be subjected to vibration applied along the axis passing through the +5 and -3 graduations of the dial. The test apparatus used shall be as specified in 4.5.4.1. The accelerometer shall be tested at each frequency and amplitude listed in TABLE II. During the test, total excursion of the pointer movement shall be recorded at each frequency stage of the test. The recorded value shall not exceed the maximum and minimum values listed in TABLE II.

4.5.4.1 Test apparatus. The equipment shall impose linear sinusoidal motion to the accelerometer and shall be capable of producing the frequencies and amplitude specified in TABLE II. The acceleration vector of any harmonic measured on the test table shall not exceed 10 percent of the fundamental acceleration vector.

4.5.5 Return mechanism operation. The accelerometer shall be placed in operating position with the pointer deflected to the maximum indication, the reset knob shall then be operated to return the pointer to zero. The force applied to the reset knob shall be between 3 to 8 pounds.

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4.5.5.1 Locking mechanism operation. The accelerometer shall be locked with the auxiliary pointers approximately 3.0g from the main pointer, the readings noted, and the accelerometer shall then be subjected to the test specified in 4.6.11. The main pointer shall not have changed more than 0.2g. The locking mechanism shall then be unlocked.

TABLE II. Response characteristic test settings and tolerances.

Impressed frequency (Hz.)	Double Amplitude (inches)	*Indicated acceleration "g"	
		Minimum	Maximum
5	2.0	4.8	5.8
6	1.5	4.8	6.3
7	1.25	4.9	7.2
8	0.75	3.4	5.6
9	0.50	2.6	4.7
10	0.50	2.8	5.8
11	0.50	0	6.7
12	0.50	0	7.5
13	0.375	0	6.2
14	0.375	0	6.8
15	0.375	0	7.1
16	0.25	0	5.0
18	0.1875	0	3.9
20	0.1875	0	3.8

NOTE: Indicated acceleration is the positive acceleration reading plus the magnitude of the negative acceleration reading.

4.5.5.2 Force on knob. A force of 15 +1 pounds shall be applied perpendicular to the shaft at the extreme outward end of the setting knob. When the force is removed, the shaft and knob shall return to the original position without permanent deformation.

4.5.5.3 Locking mechanism operation (individual test).

a. Manually shake the accelerometer to display the auxiliary pointer to at least +4g and -4g.

b. Lock the accelerometer and shake the accelerometer to the approximate same level with auxiliary pointers still in position as in "a" above. Verify that the mechanism is locked which will be indicated by the main pointer not moving during the shaking motion.

4.5.6 Low-temperature operation. The accelerometer shall be placed in a chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours. Upon completion of the test period and while still at the test temperature, the accelerometer shall be subjected to and pass the test specified in 4.5.2.1.

4.5.7 High temperature operation. The accelerometer shall be placed in a chamber and maintained at a temperature of 71° ±2°C for a period of 4 hours. Upon completion of the test period and while at the test temperature, the accelerometer shall be subjected to and pass the test specified in 4.5.2.1.

4.5.8 Magnetic effect. The accelerometer shall be held in various positions with the nearest part 5 inches from the magnetically East or West of the center of a short bar magnet compass. The horizontal magnetic field intensity shall be 0.17 to 9.19 oersted. The compass deflection for any position of the accelerometer shall not exceed 1 degree.

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4.5.9 Vibration error. The accelerometer shall be mounted on a suitable vibration stand with the dial in a vertical plane and with the graduation +5 at the top. The auxiliary pointers shall be displaced approximately 0.2g from the main pointer and the readings noted. The reset mechanism shall not be operated until the completion of the test. For a total of 3 hours the accelerometer shall be vibrated at the specified frequencies at a double amplitude between 0.018 and 0.020 inch, and the pointer oscillation shall be noted. The frequency shall be varied from 500 to 3,000 cpm in a period of not more than 60 minutes. The natural frequency of the seismic system shall be noted during the frequency variation between 500 to 3,000 cpm. For the remainder of the 3-hour test period, the accelerometer shall be vibrated at a frequency of 1,800 cpm. At the end of the test period, the indications of the auxiliary pointer shall be read before operating the resetting mechanism. At frequencies below 1,200 cpm, the oscillation of the main pointer shall not exceed 0.4g and at frequencies over 1,200 cpm, the oscillation shall not exceed 0.6g. After vibration, the indications of the auxiliary pointers shall not be changed by more than 0.2g. The accelerometer shall suffer no damage as a result of this test.

4.5.10 Vibration endurance. The accelerometer shall be mounted as specified for the vibration error test and the auxiliary pointers displaced approximately 2g from the main pointer. The indications of the auxiliary pointers should be noted and the reset mechanism shall not be operated until the completion of the test. The accelerometer shall be subject to vibration at a double amplitude between 0.018 and 0.20 inch, in the frequency range of 500 to 3,000 cpm, preferably at the frequency at which maximum pointer oscillation is obtained, for a period of 50 hours. Oscillation of the main point shall not exceed 0.8g during or after the test. At the completion of the test, the indications of the auxiliary pointers shall not be changed by more than 0.4g. The accelerometers shall then be subjected to the test specified in 4.5.2.1 The dynamic scale error shall not exceed 0.2g.

4.5.11 Shock. The accelerometer shall be mounted in the normal operating position with the mechanism locked and subject to a shock test in accordance with MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2. The stock pulse shall be half sine wave form with a peak value of 15 g's for a time duration of 11 milliseconds. After the completion of the cycle, the units shall be examined for damage and then subjected to the tests specified in 4.5.2.1.

4.5.12 Mounting lugs. The instrument case, with mechanism and cover glass removed, shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs received no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied along the axis of the mounting hold and toward the front of the case for a period of 1 minute. There shall be no fracture.

4.5.13 High-temperature exposure. The accelerometer shall be subjected to a temperature of 71° ±2° C for a period of 24 hours. After return to normal temperature, the accelerometer shall be subjected to the test specified in 4.5.2.1. There shall be no damage to the accelerometer, and the dynamic scale error shall not exceed 0.2g.

4.5.14 Low-temperature exposure. The accelerometer shall be maintained at a temperature of -65° ±2°C for a period of 48 hours. The temperature shall then be changed to -54° ±2°C and maintained for an additional 24-hour period. After return to normal temperature, the accelerometer shall be subjected to the test specified in 4.5.2. There shall be no damage to the accelerometer, and the static scale error shall not exceed 0.2g.

4.5.15 Temperature shock. The accelerometer shall be subjected to a temperature shock test in accordance with MIL-STD-810, method 503.2, procedure I. At the completion of this test, the accelerometer shall be subjected to and shall meet the requirements specified in 4.4.1.

4.5.16 Humidity. The accelerometer shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.2, procedure I. At the completion of this test, the individual tests (4.4.1) shall be repeated. No damage shall result from this test and the tolerances specified for the individual tests shall apply.

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4.5.17 Salt fog. The accelerometer shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509.1, procedure I, except for a period of 50 hours. At the end of the 50-hour period, the individual tests (4.4.1) shall be repeated. No damage shall result from this test, and the tolerances specified for the individual tests shall apply.

4.5.18 Fungus. The accelerometer shall be subjected to a fungus test in accordance with MIL-STD-810, method 508.3, except for a period of 14 days. At the end of this test, the accelerometer shall be carefully examined to ascertain that no fungus growth has occurred that would affect subsequent operation.

4.5.19 Leakage (applicable to ABU-4B/A only). The accelerometer shall be tested for leaks using a mass spectrometer type of helium leak detector. The detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.5.20 Fogging (applicable to ABU-4B/A only). The accelerometer shall be energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period, and while at the high temperature, an ice cube shall be rubbed on the face of the accelerometer glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used), and the accelerometer inspected for evidence of water and oil fog. Any evidence of fogging shall be cause for rejection.

4.5.21 Internal examination. The case of the accelerometer shall be opened and the mechanism shall be examined. Any deterioration or damage which could in any manner prevent the accelerometer from meeting functional operation requirements during service life shall be cause for rejection.

4.5.22 Lighting tests. The accelerometer shall be subjected to all the lighting tests specified in MIL-L-25467 and MIL-L-27160, as applicable.

4.6 Reliability

4.6.1 First article reliability. The six samples specified in 4.3.3 shall be subjected to a reliability test in accordance with MIL-HDBK-781, test plan III D, to demonstrate a minimum acceptable mean time between failures (MTBF) of 1000 hours. During the operative periods at approximately every 15 seconds, the accelerometers shall be subjected to an impact of sufficient magnitude to register alternately, at least, -3 and +5g units, with the auxiliary pointers reset by the PUSH TO SET knob every 2 minutes. Upon completion of each 150 operating hours the instrument shall be subjected to and shall meet the requirements of 4.5.2, 4.5.2.1, 4.5.4, and 4.5.5. A failure has occurred whenever a unit fails any test.

4.6.2 Production reliability tests. Samples of the accelerometer shall be tested as specified in 4.6.1, except that the accept-reject criteria for test pan V-D of MIL-HDBK-781 shall be applied.

4.6.3 Longevity test. Longevity test shall be performed on two accelerometer while cycling each accelerometer for 3000 hours in accordance with 4.6.1. At 200 hours operating time intervals, each accelerometer shall be subjected to and shall meet the requirement during a reliability test may be applied toward the longevity test when the same sample units are used.

4.7 Inspection and test. Test of methods of preservation and packaging shall be accomplished in accordance with MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Packaging. Packaging requirements shall be as specified in the contract or order (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The accelerometers covered by this specification are intended for use on aircraft to indicate vertical acceleration in maneuvers and in rough air.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Type of accelerometer required (see 1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue to individual documents referenced (2.2).

d. Where the first article inspection samples should be sent and activity responsible for testing (see 4.3.1).

e. Whether sampling plan B and C are to be omitted (see 4.4.2.2 and 4.4.2.3).

6.2.1 Contract data requirements. Data specified in 4.4.2 will be listed directly on a DD form 1423 incorporated into the contract.

6.3 First article provisions. It is understood that the accelerometer supplied under contract or order will be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and methods of manufacture. Deviation from the standards of the preproduction sample will be made only by the procuring activity. Evidence of unauthorized change will constitute cause for rejection.

6.4 Definitions

6.4.1 Amplitude. Whenever the word "amplitude" is specified, it means the extent of motion as measured from mean to an extreme position.

6.4.2 "G" unit. The "g" is defined as an acceleration of 32.2 feet per second per second.

6.4.3 Tapping. The phrase "tapped lightly" is defined as meaning the application of light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.4.4. Scale error. Scale error is defined as the difference between the indication of the accelerometer and the corresponding calibrated standard.

6.5 Subject term (key words) listing.

Model ABU-4A/A and ABU-4B/A
Vertical Accelerometer

6.6 International standardization agreement. Certain provisions (3.5.1, 3.5.5, 3.5.8, 3.5.9, and FIGURE 1) of this specification are subject of international standardization agreement ASCC AIR STD 10/4A and NAT-STD-3330. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization office, if required.

6.7. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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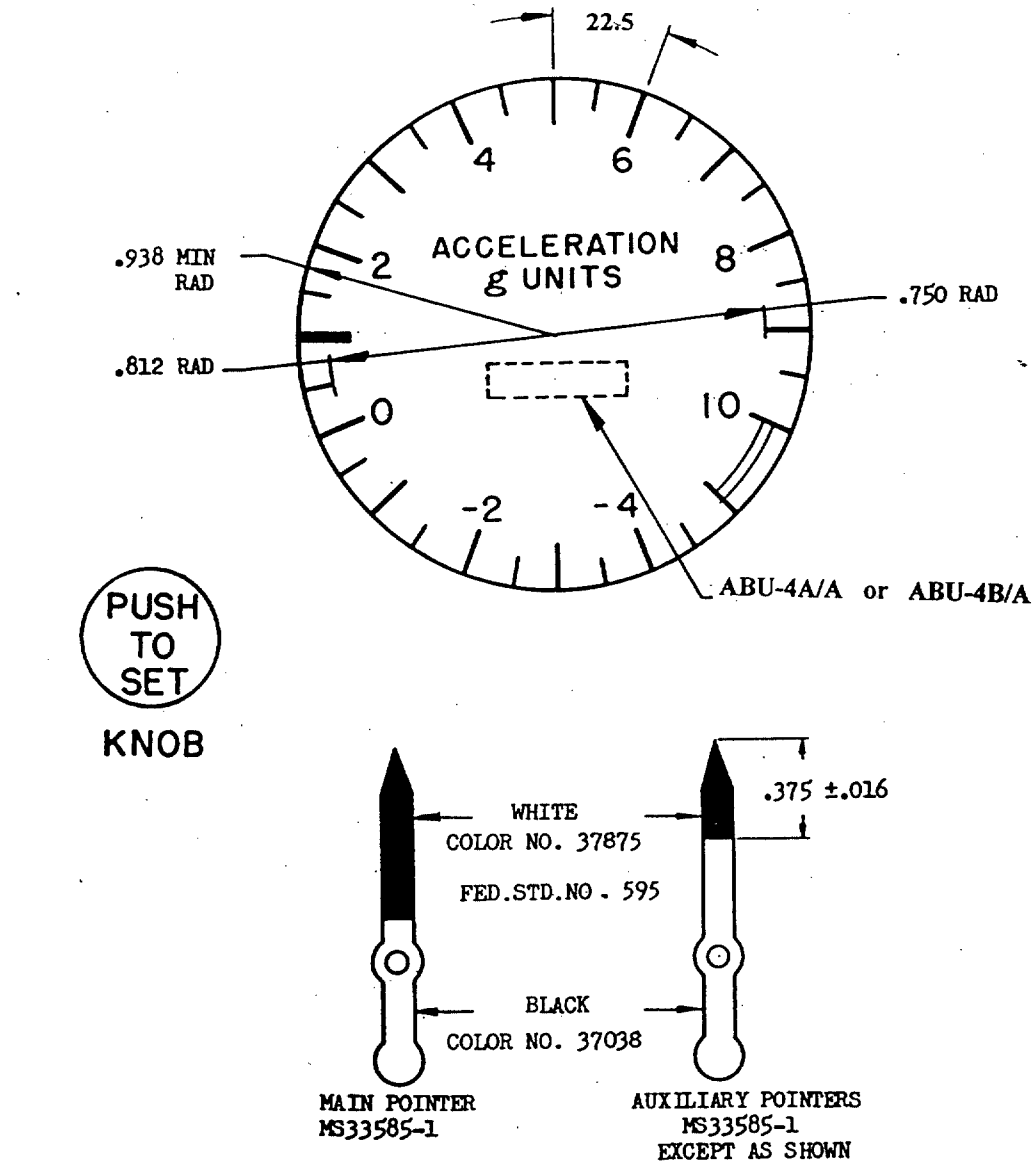


Figure 1. Dial and Pointers.

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ACCELLOMETER,AIRCRAFT RANGE -5 TO 10 G UNITS, INTEGRALLY LIGHTED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

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